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SEMESTER 1 EXAMINATIONS 2023/2024

MODULE: CA320 - Computability and Complexity

PROGRAMME(S):

COMSCI	BSc in Computer Science
CASE	BSc in Computer Applications (Sft.Eng.)
ECSAO	Study Abroad (Engineering & Computing)
ECSA	Study Abroad (Engineering & Computing)

YEAR OF STUDY: 3, O, X

EXAMINERS: Dr. David Sinclair (Ext. 5510)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer 4 Questions.
All questions carry equal marks.

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The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

Requirements for this paper (Please mark (X) as appropriate)

☐	*Log Tables*	☐	*Thermodynamic Tables*
☐	*Graph Paper*	☐	*Actuarial Tables*
☐	*Dictionaries*	☐	*MCQ Only - Do not publish*
☐	*Statistical Tables*	☐	*Attached Answer Sheet*

Section A

QUESTION 1 ***[Total marks: 25]***

1(a) [5 Marks]

Using examples, show how the `let` and `where` constructs are used in Haskell.

1(b) [5 Marks]

Using an example, illustrate how the `case` statement can be used in conjunction with pattern matching in Haskell.

1(c) [15 Marks]

Representing sets as lists, write a Haskell functions that implements set union and set difference. Remember sets do not have duplicates. The function should use class constraints to ensure that it will work for sets of integers, floats, and characters.

[End Question 1]

QUESTION 2 ***[Total marks: 25]***

2(a) [5 Marks]

Write a Haskell data structure that represents an *ordered binary tree*.

2(b) [10 Marks]

Write a Haskell function that inserts elements into the correct position in an ordered binary tree.

2(c) [10 Marks]

Write a Haskell function that calculates the length of the shortest path from the root of a binary tree to a leaf node.

[End Question 2]

Section B

QUESTION 3 ***[Total marks: 25]***

3(a) [5 Marks]

Define *regular expressions*.

3(b) [10 Marks]

Formally define the machine that recognises languages generated by regular expressions.

3(c) [10 Marks]

State the *Pumping Lemma for Regular Languages* and what is its significance?

[End Question 3]

QUESTION 4 ***[Total marks: 25]***

4(a) [5 Marks]

In the study of Computability & Complexity, why do we focus on languages?

4(b) [5 Marks]

Describe the terms *computability* and *complexity*. How do these terms relate to one another?

4(c) [15 Marks]

What is a *Universal Turing Machine*? What is its significance?

[End Question 4]

QUESTION 5 ***[Total marks: 25]***

5(a) [5 Marks]

Formally describe the Big-O Notation. Rank five typical Big-O categories from “quickest” to “slowest”.

5(b) [5 Marks]

In terms of Complexity, define NP-completeness.

5(c) [15 Marks]

The Cook-Levin Theorem states that *Satisfiability* is NP-complete. Give an outline of the proof of the Cook-Levin Theorem.

[End Question 5]

[END OF EXAM]